



# **Basic Emergency Care**

## **Loss of consciousness**

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# What is Loss of Consciousness (LOC)?

- Temporary loss of awareness and responsiveness
  - A symptom, not a diagnosis
  - Treat as potentially serious

# Pathophysiology of Consciousness

- Requires intact brainstem (RAS) + cerebral cortex
- Needs oxygen, glucose & cerebral perfusion
- 6–10 seconds without blood flow → LOC
- Prolonged hypoxia → brain injury

# Major Causes

- Seizure
- Stroke
- Hypoglycemia
- Cardiac causes
- Hypoxia / Drowning

# Causes of LOC

- Neurological: seizure, stroke, hemorrhage
- Metabolic: hypoglycemia, DKA, electrolytes
- Cardiac: arrhythmia, MI, shock
- Respiratory: hypoxia, drowning
- Toxic: overdose, alcohol, CO

# Initial Approach – DRABC

- D – Danger
- R – Response (AVPU)
- A – Airway
- B – Breathing
- C – Circulation

# Glasgow Coma Scale (GCS)

- Eye (4) + Verbal (5) + Motor (6) = 15
- $GCS \leq 8 \rightarrow$  airway protection needed
- Trend scores over time

# Seizure

- Abnormal brain electrical activity
  - May cause jerking movements and LOC
  - Post-ictal confusion common



# Seizure – Clinical Features

- Aura may precede event
- Tonic phase → stiffness
- Clonic phase → jerking
- Post-ictal confusion common

# Seizure – First Aid

- Protect from injury
  - Do NOT restrain
  - Do NOT put objects in mouth
  - Place in recovery position after seizure
  - Call emergency if >5 minutes

# Stroke

- Sudden loss of brain blood supply
  - Medical emergency

# Stroke – Additional Points

- Ischemic vs hemorrhagic
- Sudden severe headache → possible hemorrhage
- Time is brain
- Urgent CT required

# FAST Assessment

- Face drooping
  - Arm weakness
  - Speech difficulty
  - Time to call emergency

# Hypoglycemia

- Low blood glucose
  - Sweating, confusion
  - May cause LOC

# Hypoglycemia – Clinical Clues

- Sweating, tremor, palpitations
- Confusion, slurred speech
- Seizure or LOC if severe
- Always check glucose

# Hypoglycemia – What To Do

- If conscious: give sugary drink
  - Recheck after 15 minutes
  - If unconscious: do NOT give orally
  - Call emergency help



# DKA

- Serious diabetic complication
  - High glucose + dehydration
  - Requires hospital treatment

# DKA – Key Features

- Hyperglycemia + ketosis + acidosis
- Polyuria, polydipsia, vomiting
- Kussmaul breathing
- Needs IV fluids + insulin

# Drowning

- Respiratory impairment due to water submersion
  - Can cause hypoxia and LOC
  - Remove from water safely
  - Start CPR if not breathing normally

# Drowning – Management

- Remove safely from water
- Check breathing & pulse
- Start CPR if needed
- Monitor for respiratory complications

# Key Take-Home Messages

- Check glucose in unconscious patient
  - Use recovery position when appropriate
  - Recognize stroke early (FAST)
  - Early emergency call saves lives